SPE-227302-MS

An Innovative Closed-Loop Geothermal Well Solution for Large-Scale District Cooling Projects

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Typical geothermal district cooling projects are characterized by high upfront investment in drilling and have geological requirements which pose geographical limitations as to where a project can be executed. This paper introduces a new closed-loop horizontal geothermal well technology that helps harness the abundantly available lower enthalpy geothermal energy for direct use in district cooling projects at most locations using an absorption chiller.

The new technology utilizes a single closed-loop horizontal well to replace the typical two wells in a standard hydrothermal solution (one producer and one injector) with specific geological requirements. This is achieved by using vacuumized pipe-in-pipe completion installed in a closed-loop horizontal geothermal well. By injecting return water from the district cooling network into the well in the annulus between the pipe-in-pipe completion and the outer casing/liner section along the geological formations in the horizontal section of the well, this working fluid is heated to reach the temperature of the geological formations. It then flows back to the surface inside the vacuumized pipe-in-pipe completion with minimal heat loss for direct use in district cooling grids by using an absorption chiller.

Implementation of the technology shows significant upscaling potential both geographically, regarding the number of wells that can be completed, and the thermal energy output achieved. The well design depends on the local geothermal temperature gradient and the thermal conductivity needed to achieve the required thermal energy output. The very limited geological requirements enable harvesting the abundantly available geothermal energy in most places. As it is a closed-loop system, there is no need for reservoir stimulation, continuous depletion of underground water aquifers, well interventions, or downhole equipment maintenance, which minimizes operational costs. Simulations showed that the execution of 10 wells with a depth of 4 km each and 3-6 km horizontal sections could deliver about 20 MW of thermal energy with 90 °C – 100 °C fluids at the surface which will be diverted to an absorption chiller to provide 16 MW thermal plus with water at 6 °C ready for direct use in district cooling applications. This entire system is estimated

to reliably work for 50+ years with minimal need for maintenance and almost no requirement for any well interventions. This paper introduces a thorough review of the breakthrough innovation in geothermal district cooling by introducing a closed-loop horizontal single-well system that can add significant value and, at the same time, mitigate the issues with conventional hydrothermal solutions.

Introduction

The Evolution of Geothermal Energy

For thousands of years, geothermal energy has been among the oldest energy sources ever utilized by humans, and in different applications. Hot springs were used by the Maoris in New Zealand and native Americans for medical purposes and cooking for thousands of years. Ancient nations such as the Romans and the Greeks had geothermally heated spas, and they also used the hot spas to treat diseases such as skin and eye diseases. People in Pompeii used the hot springs to heat buildings. In modern days, the first commercial use of geothermal energy was in the 18th century (1892) in district heating in Boise, Idaho, USA. In the 19th century, the development of thermodynamics eventually led to new applications of geothermal energy, such as electricity generation from geothermal heat by transforming the hot steam energy into mechanical force and then into electricity (Allahvirdizadeh, 2020). Many other applications and utilizations for geothermal energy have matured since then, as reflected in Fig. 1, with a Lindal diagram illustrating the potential utilizations of geothermal fluids at various temperatures.

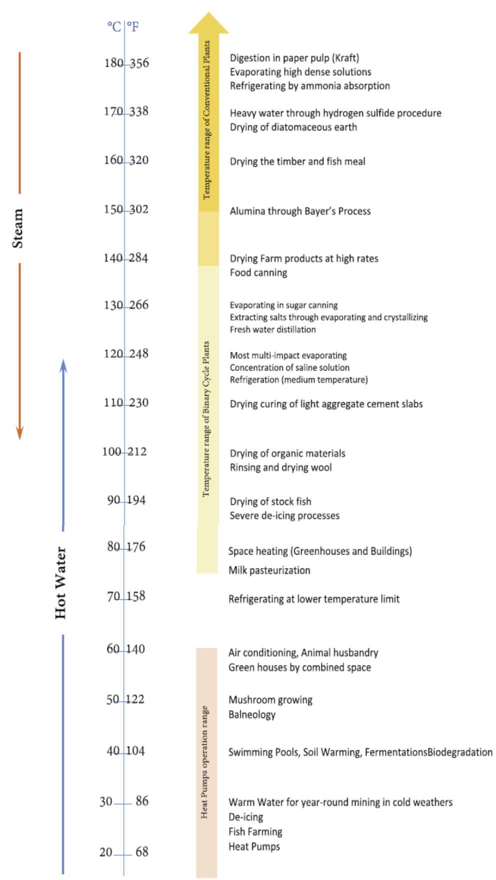

Figure 1—Modified Lindal diagram (Allahvirdizadeh, 2020).

Most of the direct heat use applications (Fig. 1) are widely used in many parts of the world, with district heating/cooling, space heating, agricultural and greenhouse applications, as well as industrial applications being the most common applications of geothermal energy at low to medium enthalpy levels (Allahvirdizadeh, 2020).

Geothermal Energy and Greenhouse Gas Emissions

Global warming, resulting from the continuously increasing Greenhouse Gas (GHG) concentration in the Earth's atmosphere, has long been recognized as the main reason behind the more frequent and severe natural catastrophic events that have been actively striking our planet. Studies show that sea level rise has been increasing at an accelerating pace and by an estimated four to five inches from 1900 to 1990, from 1990 to 2015 by three inches, while the current rate of sea level rise is one-eighth of an inch per year (NOAA, 2021). Additionally, an increase of 1 °C in the atmospheric temperature will result in an increase in the moisture level by 7% which can lead to higher precipitation that can produce larger floods, stronger hurricanes, and snowstorms (Gibbens, 2024).

Moving away from fossil fuel consumption to a renewable energy mix is one of the main pillars for reducing global GHG emissions, which is becoming more crucial for humanity. A globally increasing intensity in harnessing high-enthalpy geothermal resources for generating electricity is represented in Fig. 2, which summarizes the installed capacity data for the top 10 countries over five periods starting from the year 1945 until 2020.

	Period	1st		2nd		3rd		4th		5th		Total	
	Interval (years)	1945–1957 (13)		1958–1976 (18)		1977–1991 (15)		1992–2002 (11)		2003–2020 (18)		1945–2020 (95)	
Group	Country (year of first commercial plant)	Addition (MW)	MW/Yr	Addition (MW)	MW/Yr	Addition (MW)	MW/Yr	Addition (MW)	MW/Yr	Addition (MW)	MW/Yr	Addition (MW)	MW/Yr
1	United States (1960)	0.0	0.0	330.0	17.4	2,219.3	148.0	266.3	24.2	860.4	47.8	3,676.0	60.3
2	Philippines (1977)	0.0	0.0	0.0	0.0	888.0	59.2	1,043.0	94.8	-2.9	-0.2	1,928.1	43.8
	Mexico (1973)	0.0	0.0	37.5	2.0	582.5	38.8	223.0	20.3	120.0	6.7	963.0	20.1
3	New Zealand (1958)	0.0	0.0	192.0	10.1	69.0	4.6	104.4	9.5	639.6	35.5	1,005.0	16.0
	Italy (1913)	47.0	3.6	87.5	4.6	294.0	19.6	356.5	32.4	159.0	8.8	944.0	12.4
	Iceland (1969)	0.0	0.0	3.0	0.2	41.8	2.8	157.2	14.3	553.0	30.7	755.0	14.5
	Japan (1966)	0.0	0.0	60.5	3.2	185.1	12.3	265.6	24.1	89.8	5.0	601.0	11.6
4	Indonesia (1979)	0.0	0.0	0.0	0.0	140.0	9.3	645.0	58.6	1,345.5	74.8	2,130.5	38.7
	Türkiye (1984)	0.0	0.0	0.0	0.0	21.0	1.4	-3.0	-0.3	1,531.0	85.1	1,549.0	41.9
	Kenya (1985)	0.0	0.0	0.0	0.0	45.0	3.0	13.0	1.2	803.0	44.6	861.0	23.9
Top 10		47.0	3.6	710.5	37.4	4,485.7	299.0	3,071.0	279.2	6,098.4	338.8	14,412.6	189.6
Global		47.0	3.6	776.5	40.9	4,629.2	308.6	3,321.2	301.9	6,669.9	370.6	15,443.8	203.2

Figure 2—Summary of installed electricity generation capacity for the top 10 countries (Ediger and Akar, 2023).

While each period in Fig. 2 represents a milestone in the geothermal electricity generation global development, the countries were grouped based on the total addition and annual average addition within the defined period (Ediger and Akar, 2023). Fig. 3 illustrates the year 2020 installed capacity per country and demonstrates that the USA had the largest installed capacity, 3,794 MW, by the year 2020. Indonesia, Philippines, Turkey, and New Zealand are also among the top five countries with geothermal installed electricity capacity.

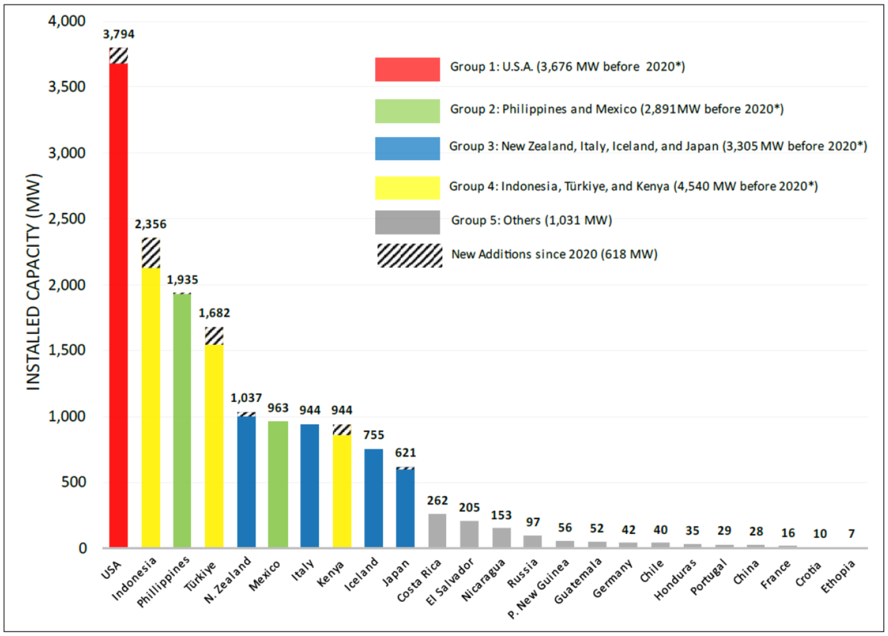

Figure 3—Installed capacity per country for the year 2020 (Ediger and Akar, 2023).

Geothermal Energy for District Heating Applications

District heating is a system for distributing heated fluids generated in a centralized plant through a network of insulated pipes for residential or commercial heating requirements. The heat source is typically obtained from plants feeding on a wide range of energy sources, including fossil fuels, biomass, boiler stations, geothermal heat, heat pumps, solar heat, factories' waste heat, or nuclear power electricity generation. Geothermal direct-use district heating systems, in general, consist of three major parts: A production well to produce the hot water delivered at the surface; a mechanical system, consisting of piping, a heat exchanger, and controls, to deliver and exchange the heat to the targeted space; and water disposal system, which could be an injection well, storage pond, or river, for disposing of the produced cooled geothermal fluid.

Since its initial use in Boise, Idaho, geothermal district heating systems have been increasingly adopted globally, supported by technological advancements and declining costs. Fig. 4 is a demonstration of a district heating system installed and in operation since 2013 in Poddebice, Poland, which has a heating capacity of 4-5 MW thermal (MWth), with an additional peak load boiler capacity of around 2 MWth (Kępińska et al, 2017). This system consists of a single production well (GT-2) capable of producing 68-69 °C water at a production rate ranging from 190 m^3/hr to 250 m^3/hr, feeding its supplies into two 5 MWth heat exchangers within the Heat Central and from there to the peak load boilers, to be used during the peak season by burning fossil fuel or biomass to heat the water and then through the pipelines to the distribution network. The heat users could either be direct users of heat or they could be equipped with heat exchange pumps, depending on the nature of their needs. From the heat users, the water will then be circulated back to the Heat Central through three circulation pumps with a 4-5 MWth heat capacity and to the water-cooling pond.

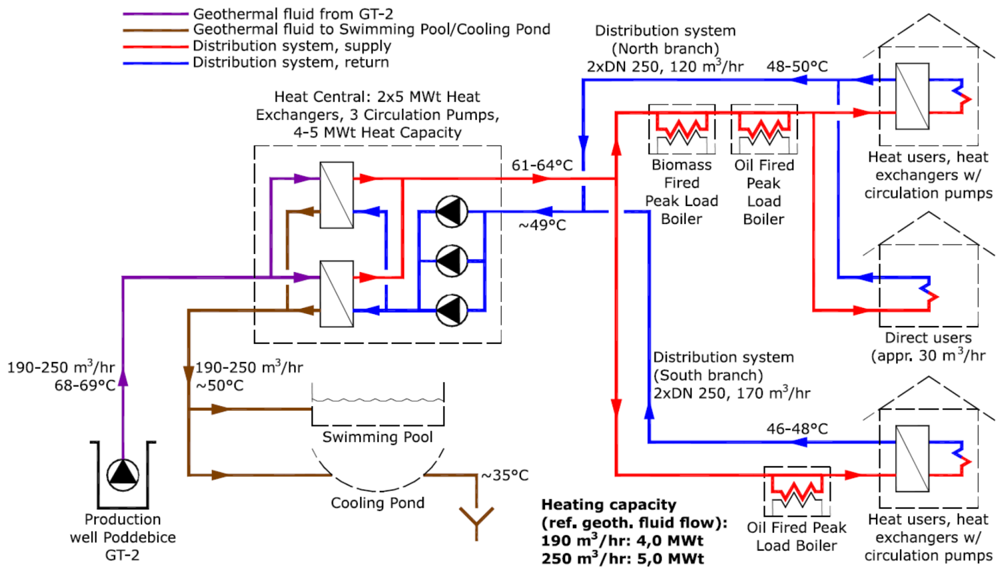

Figure 4—Geothermal district heating system diagram in Poddebice, Poland (Kępińska et al, 2017).

Geothermal Energy for District Cooling Applications

Geothermal energy application in district cooling is relatively more recent, but it is anticipated to be more widespread, especially in warm and humid countries, driven by the global demand for air-conditioning, which is projected to triple within the next 30 years. When used effectively, district cooling is more energy-efficient than conventional cooling methods, delivers greater cost benefits to consumers, and significantly reduces peak power demand. As stated by Sarraf et al. (2019), "Using district cooling for future additional cooling demand in developing countries could lead to over US$1 trillion in energy savings worldwide through 2035". It also aligns seamlessly with solar energy and other emerging renewable technologies, enhancing the overall sustainability of the energy supplies. Gulf Cooperation Council (GCC) countries, being among the first to successfully implement large-scale geothermal district cooling projects, can provide perfect use-case scenarios for evaluation and optimization of future geothermal district cooling project designs (Sarraf et al, 2019). As shown in Figure 5, district cooling systems rely on a central chiller plant to circulate chilled water through an insulated pipeline network to several buildings. Within each building, a heat exchanger transfers the cooling effect from the chilled water to the building's air conditioning system. The system operates in a closed loop, where the chilled water, after finishing cooling the buildings, returns to the central plant to be re-chilled and recirculated (Sarraf et al, 2019). The most common cooling technology that relies on heat to produce the chilled water needed for district cooling is called the absorption chiller.

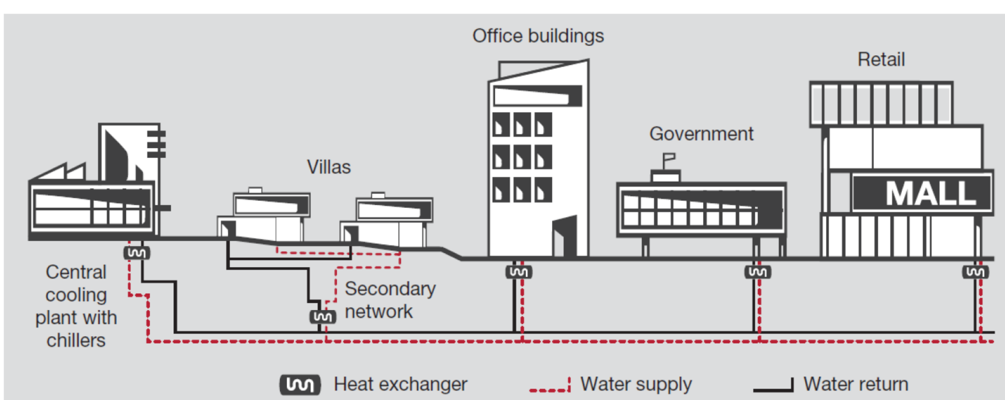

Figure 5—A typical district cooling system (Sarraf et al, 2019).

Single-Absorption Chillers District Cooling Systems

Single-effect absorption chillers are energy-efficient cooling systems that involve thermodynamic cycles using a heat source to drive the refrigeration process. The primary components in a typical single-effect absorption chiller system include a generator, absorber, condenser, evaporator, and an expansion valve. It uses a lithium bromide solution as absorber passing through the generator, heat exchanger, and the absorber, and uses water as refrigerant circulating through the condenser and the evaporator, and in between the two, through the expansion valve (El Haj Assad et al, 2021). As shown in Fig. 6, the hot water coming from the production well is pumped through pipes into the generator to heat the lithium bromide solution, which will be separated into water vapor and heavy lithium bromide solution precipitating at the bottom of the generator. The heavy solution will circulate to the heat exchanger and then to the throttling valve to have a low pressure before entering the absorber. In the absorber, the heavy solution will be sprayed in the absorber and mixed with the water before being pumped again into the generator, passing through the heat exchanger, to complete the closed loop. Back to the water vapor formed in the generator, it will circulate to the condenser, where it will be cooled down by the cold pipes carrying cold water from the absorber to the condenser in a closed loop. The water vapor will condense and drop into the collection trays and circulate through the expansion valve to reduce the liquid pressure to near-vacuum conditions. This sudden change in pressure will cause the water entering the evaporator to flash and drop in temperature to very cold water, having a temperature of 4.5 °C. This chilled water will be used for district cooling by passing chilled water pipes in the evaporator, coming from the end user buildings, bringing heat from the buildings. The water pipes will pass through the evaporator and be covered by a thin layer of chilled water, which will cool down the water inside the pipes. The cooled water will then be circulated to the end user building for cooling purposes.

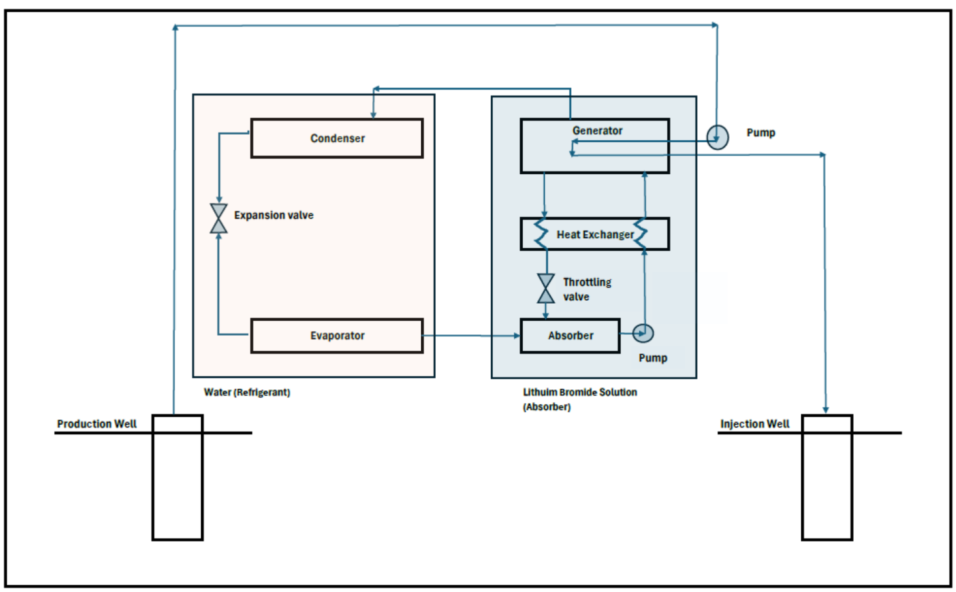

Figure 6—A Schematic diagram for a single-effect geothermal district cooling system (Modified after El Haj Assad et al, 2021).

Absorption chillers offer several advantages over traditional mechanical refrigeration systems and other chillers. In addition to their capacity to use high-temperature geothermal fluids, single-effect absorption chillers can utilize low-grade heat sources, such as waste heat, solar and thermal energy, to generate cooling, significantly reducing electricity consumption and operating costs. These systems operate with natural refrigerants like water and lithium bromide, which are non-ozone-depleting and have low global warming potential, making them environmentally friendly. Additionally, the absence of mechanical compressors results in quieter operation, making absorption chillers suitable for noise-sensitive environments. With fewer moving parts, these chillers experience less wear and tear, leading to lower maintenance requirements and longer lifespans. While the initial investment may be higher, the long-term energy savings and reduced maintenance costs make absorption chillers a cost-effective and sustainable choice.

Electricity Generation – Geothermal Power Plant

The district cooling system can take water having a temperature of 90 °C to 100 °C as an intake, but if the production well produces water that has higher temperatures, about 250 °C, then it can be used for generating electricity and then cycled to the single-effect district cooling system. In this scenario, the produced water will flash into vapor at the surface, creating huge momentum and passing through the gas turbine to generate electricity. Once this cycle is finished, the vapor will be circulated to a condenser to drop back to a liquid, having a high enough temperature, which will be sent to the generator within the single-effect geothermal district cooling system to complete the cooling cycle (El Haj Assad et al, 2021).

Absorption Chillers Cooling Efficiency

The cooling efficiency of a single-effect geothermal district cooling system is a function of several key parameters. Most importantly, the temperature of the inlet fluid, which is injected into the generator to heat the lithium bromide solution. The higher the fluid inlet temperature, the higher the Coefficient of Performance (COP), and the higher the cooling load. Additionally, the heat exchanger's size and effectiveness play important roles in the efficiency of the single-effect geothermal district cooling system. While the heat exchanger effectiveness influences the heat transfer load of the generator and the absorber in a direct relationship, the heat exchanger size results in an exponential increase in the cooling capacity. Both the COP and the cooling load also increase as the size of the evaporator increases (El Haj Assad et al, 2021).

A Technology Frontier Solution: Single Closed-loop Geothermal Well for District Cooling

In a new geothermal energy application breakthrough, recent research has matured a technology that enables harnessing geothermal energy for large-scale district cooling applications using a single closed-loop geothermal well system that uses a single, relatively deep geothermal well, in contrast to the more commonly used doublet configuration, which consists of separate production and injection wells. This approach introduces significant optimizations and reduces upfront CAPEX requirements, which are usually the main obstacle to executing large-scale geothermal district cooling projects.

Technology Overview

Compared to the typically used doublet well solution for cooling applications, this technology utilizes a vacuumized pipe-in-pipe completion installed within a closed-loop horizontal geothermal well, as shown in Fig. 7.

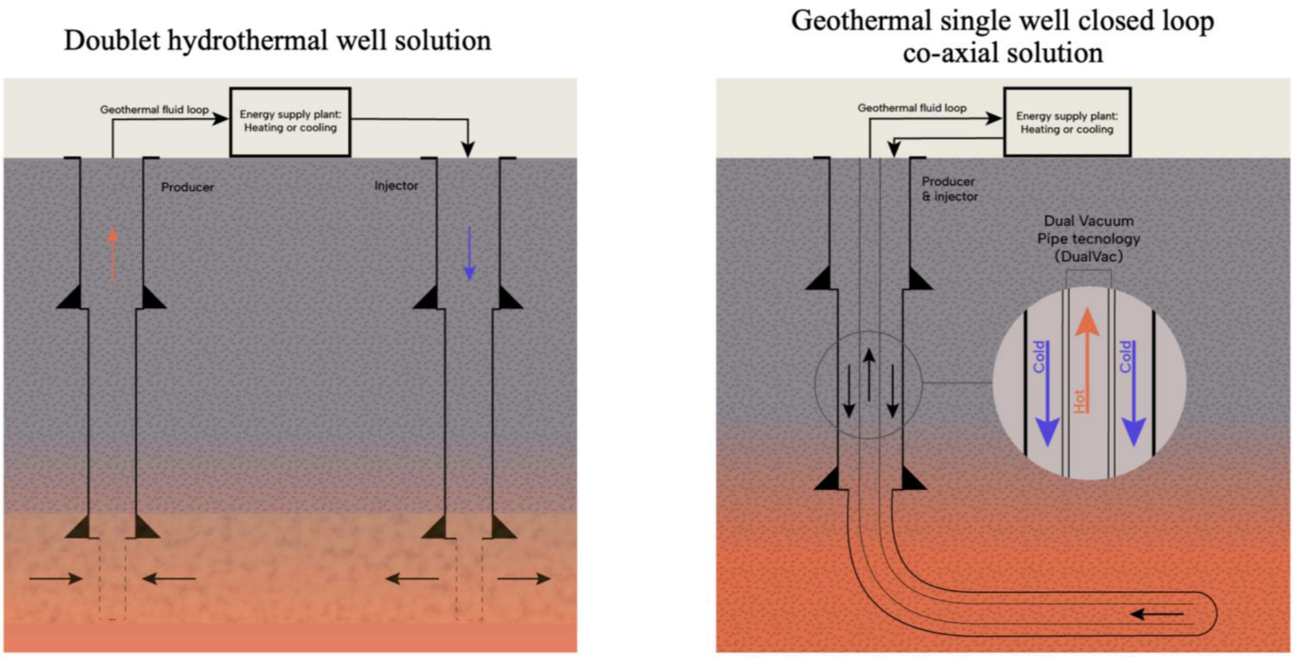

Figure 7—Schematic drawing for a geothermal closed-loop system with a single closed-loop well having vacuumized pipe-in-pipe completion (Modified from Maver and Vestavik, 2023).

For the single closed-loop well solution to work, it requires drilling the well to reach a rock formation characterized by high thermal conductivity, having a temperature ranging between 100 °C to 150 °C, hence the vertical depth of those wells can range from 3 km to 5 km. Once this rock formation is reached, a horizontal section of 3 km to 6 km is drilled into the rock formation and completed using the pipe-in-pipe completion. The key factor in determining the length of the horizontal section is the target thermal energy output from the well, so the larger the required thermal energy, the longer the horizontal section should be. Fig. 8 represents a simulation for the temperature and thermal energy output over time versus the length of the drilled horizontal section with a constant thermal conductivity.

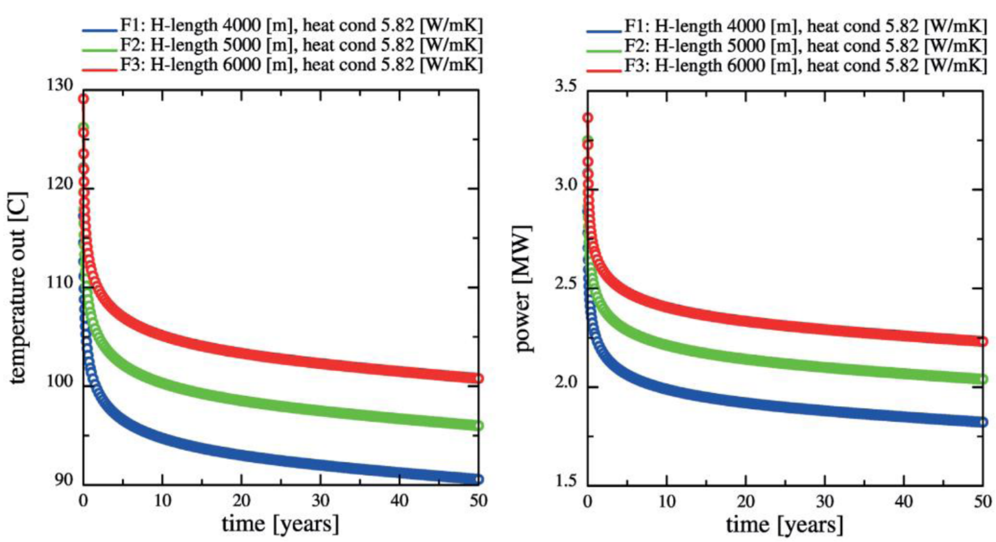

Figure 8—Example of temperature and thermal power for the return fluid vs time for a horizontal section of 4, 5 and 6 km (Maver and Vestavik, 2023).

As established by simulation, once the fluid starts circulating, the geological formation surrounding the borehole begins to cool rapidly. Within a few months, cooling extends to about 10 m from the borehole.

However, the progression of this thermal change, which can be referred to as the "cold front" slows over time. After approximately 10 years of continuous operation, the cold front advances to around 40 m away from the borehole. With stable heat extraction, the movement of the cold front continues at a progressively slower rate, following a logarithmic trend. After 50 years of production, the cold front typically reaches a distance of about 100 m from the borehole. At a horizontal well depth of 4 km, the temperature variation in the surrounding formation remains relatively small, having only a minor effect on the thermal output of the system.

Assuming a consistent production rate of 2 MWth over 50 years, the average temperature drop within the cylindrical volume of geological formation surrounding the well is estimated at approximately 8.2 °C. The greatest cooling occurs at the borehole wall near the heel, while the temperature change is least pronounced at the toe of the horizontal section. Once the well is drilled, completed, and connected to the surface facilities, the return water from the district cooling network is injected into the annular space between the inner pipe and the outer casing or liner along the horizontal section of the well. As the water flows through this section, it absorbs heat from the surrounding geological formations, reaching subsurface temperatures. The heated fluid then returns to the surface through the inner vacuumized pipe-in-pipe completion with minimal heat loss and is used directly in district cooling systems via absorption chillers (Maver and Vestavik, 2023).

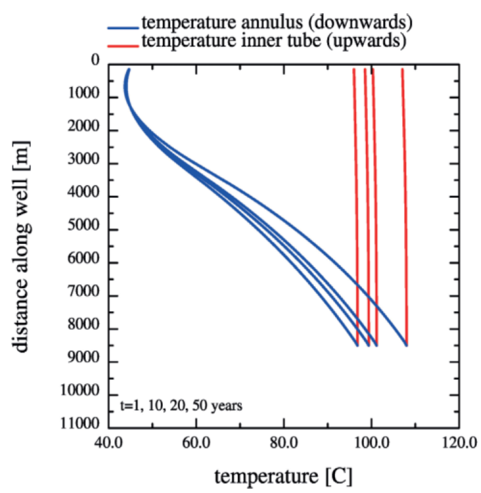

Figure 9—Modelling of the very limited heat loss in the inner pipe taking the heated fluid to the surface after 1, 5, 10 and 50 years for a 3.7 km-deep well with a 4 km horizontal section and a fluid flow rate of 6 kg/s (Maver and Vestavik, 2023).

Importance of Vacuumized Pipe-in-Pipe Completion. To complete the single closed-loop well, an outer casing/liner is installed with an inner vacuumized pipe-in-pipe completion. During installation, the vacuum pipe's seals are regularly tested. If a major leak is detected during or after installation, the completion string may need to be pulled for repairs. The entire steel assembly is sealed from oxygen and external fluids, giving it long-term durability with minimal maintenance. Without the vacuum insulation, the returning fluids suffer significant heat losses. But with the help of the installed vacuumized pipe-in-pipe completion, heat loss while the fluid travels upward is negligible, as shown in Fig. 9, using a model developed by the Institute for Energy Research (IFE) in Kjeller, Norway, and validated through comparisons with LedaFlow and FeFlow simulation models (Maver and Vestavik, 2023).

Key Technology Features. Several advantages are achieved with the implementation of the new approach of the single closed-loop well. For instance, in the doublet well option, the system's performance is highly reliant on the thickness of the reservoir formation and other key formation characteristics such as reservoir temperature, formation pressure, porosity, permeability, and geochemical properties. These factors are critical to ensuring sufficient water productivity, thermal gradient, and hydrologic connectivity between the wells to sustain continuous fluid production. On the other hand, there are very limited geological requirements in the single closed-loop well solution, which are the reservoir temperature and thermal conductivity of the geological formations especially of the horizontal well section. As this solution requires no direct interaction between the fluids inside the well and the reservoir, and as there is no reservoir water production, it practically eliminates the otherwise much-needed well maintenance and surveillance that will be required in the doublet well scenario due to corrosion, scaling, and clogging of the wells. In addition to the high maintenance cost, the doublet well scenario often requires additional hydrofracturing of the reservoir to guarantee communication between the producer and injector wells, representing another major expense item which is eliminated in the single closed-loop well solution (Maver and Vestavik, 2023).

Geological Considerations

Several geological parameters are of key importance when it comes to the successful and economic implementation of the proposed single closed-loop well solution.

Thermal Conductivity. Rock thermal conductivity is affected by a number of parameters, but most importantly by rock composition and grain-to-grain heat transfer efficiency, which in turn is affected by the pore space within the rock, the more porosity, the less effective the heat transfer, hence the lower the thermal conductivity. This is why studies show a distinct and predictable correlation between the acoustic compressional wave velocity and the rock conductivity (Pimienta et al, 2014). The rock thermal conductivity is also affected by its temperature and pressure, with a direct relationship between the pressure and rock conductivity and an inverse relationship between the temperature and rock conductivity (Maver and Vestavik, 2024).

While rock salt exhibits consistently high and predictable thermal conductivity, with values exceeding 6 W/(m·K) at 20 °C and slightly decreasing to just under 5 W/(m·K) at 160 °C (Raymond et al, 2021), thermal conductivity varies widely among other rock types. For example, dry, unconsolidated sedimentary rocks such as sands and gravels tend to have low conductivity. Moderate to high values are typical of most sedimentary and metamorphic rocks, while felsic igneous rocks can exhibit very high conductivity. Quartz-rich rocks, like sandstone, and water-saturated formations also conduct heat well (Maver and Vestavik, 2024).

Geothermal Temperature Gradient. The geothermal temperature gradient exhibits two distinct patterns depending on whether the measurements originate from oceanic or continental crust. The median geothermal gradient in oceanic regions is approximately 64 °C/km. However, for practical geothermal energy extraction, the relevant gradient is that of the continental crust, which has a lower median value of about 34 °C/km. This gradient can vary significantly depending on the geological structure and setting of the crust (Maver and Vestavik, 2024). There is a well-established correlation between crust thickness and its thermal gradient, so the thinner the crust, the higher its thermal gradient will be, especially if this crustal thinning is near an active divergent tectonic boundary (Bukhamseen et al., 2024).

Heat flux data is also found useful for identifying higher geothermal gradient areas, even though high heat flux does not always indicate a high geothermal gradient (Maver, 2025). This is mainly because the total heat flow from Earth's interior into space is approximately 44 terawatts (TW) (Johnston, 2019), and it can be broken down as follows:

- 24 TW originates from primordial heat, generated from Earth's inner mantle (Bukhamseen et al, 2024). This thermal energy can be directly linked to the thermal gradient as it emanates from the Earth's interior
- The remaining 20 TW can be a near-surface feature, and it is associated with the radiogenic decay of three key radioactive minerals primarily existing within basement rocks such as granites

Rock Hardness and Rate Of Penetration. For the implementation of the single closed-loop well three types of lithology are considered. First, and a very common lithology, is the sedimentary rocks. Both vertical and directional drilling through sedimentary rocks is supported by well-established, mature oil and gas drilling technologies, allowing for relatively high Rates Of Penetration (ROP) during drilling operations. However, drilling ROP for sedimentary rocks can vary depending on many factors, but most importantly, the rock hardness, drill bit type, drilling parameters, and bottom hole assembly (BHA). The second lithology is thick rock salt deposits or salt structures located at depths that allow the horizontal section to reach high temperatures and high thermal conductivity. Rock salt can have a thermal conductivity that is 2-4 times higher than that of non-evaporitic sediments. Drilling through salt is relatively fast, with an average ROP reaching up to 40 m/hr, which makes those targets even more attractive from an economic point of view, as it significantly reduces drilling cost. Basement rocks, such as volcanics, igneous, and metamorphic rocks, are also considered. Areas such as the Ring of Fire, where the tectonic plates are in a divergent motion, volcanic arcs, and volcanic belts, which are formed in convergent plate boundaries, all exhibit an elevated level of geothermal gradient. However, drilling through those hard and abrasive rocks, which typically requires additional bit runs, results in relatively slow drilling progress, with the highest possible average ROP of 20 m/hr due to the excessive rock hardness and shock and vibration, which disperse the drilling energy. Fig. 10 below illustrates a map showing the global distribution of sedimentary rocks thicker than 4 km, and rock salt ideal for a commercial single closed-loop well (Maver, 2025).

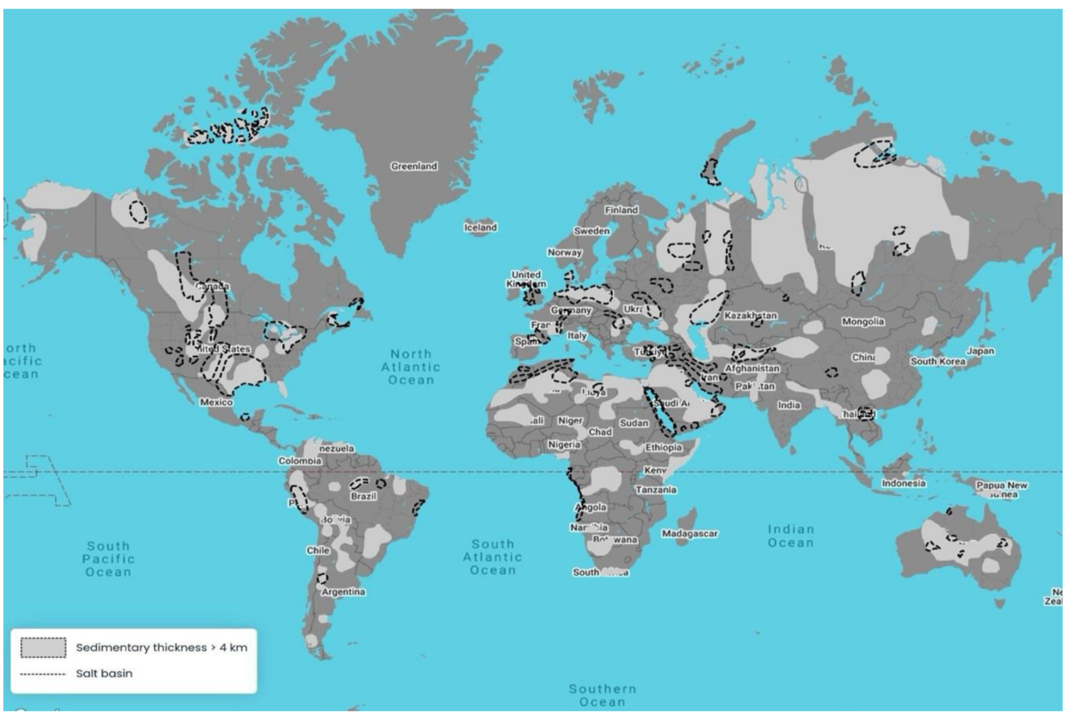

Figure 10—The global distribution of sedimentary rocks thicker than 4 km, and rock salt ideal for a commercial single closed-loop well (Maver, 2025).

Single Closed-Loop Well Heat Utilizations in A Simulated District Cooling Project

Assuming a district cooling project requiring 20 MW of input thermal power, an estimated 10 single closed-loop wells need to be drilled. Those wells will vary in their designs, depths, length of the horizontal sections, and costs, depending on their thermal properties and the lithology of the drilling targets. Table 1 below summarizes the single closed-loop well design parameters and thermal properties for drilling into three different target lithologies: sedimentary rocks, basement rocks, and rock salts. For district cooling using geothermal energy, a single-effect absorption chiller, designed for 80-120 °C water input, will be used and typically produce 6-7 °C chilled water (Maver, 2025).

Table 1—Example of well completion requirements for a 10 single closed-loop wells project with a total of 20 MWth output for three geological settings, assuming an initial 150 °C virgin formation temperature at the horizontal section at a 4.5 km vertical depth (Maver, 2025).

Lithology Thermal Conductivity	Temperature Scenario 1	Well Length Scenario 1	Temperature Scenario 2	Well Length Scenario 1
Sedimentary rocks 2.5 W/(m·K)	Long-term produced temperature: 85 °C Return temperature: 45 °C	Horizontal: 6,150 m Total: 10,650 m	Long-term produced temperature: 70 °C Return temperature: 38 °C	Horizontal: 5,000 m Total: 9,500 m
Basement (igneous and metamorphic rocks) 3.5 W/(m·K)		Horizontal: 4,400 m Total: 8,900 m		Horizontal: 3,600 m Total: 8,100 m
Rock salt 5.0 W/(m·K)		Horizontal: 3,100 m Total: 7,600 m		Horizontal: 2,500 m Total: 7,000 m

GCC District Cooling – Modelling & Scenarios

Cooling is one of the biggest challenges faced by countries in the GCC region, with summer temperatures often exceeding 50 °C. 70% of electrical consumption in Saudi Arabia is in cooling, and as Saudi Arabia strives for more energy-efficient and carbon-neutral economy, the district cooling technology gains critical importance in achieving these targets (Saudi Tabreed, 2025), having district cooling systems that can provide cooling solutions to commercial buildings, hotels, apartment blocks, shopping malls etc. (Zafar, 2025). The GCC region air conditioner market is a critical component of the region's infrastructure, driven by extreme climatic conditions and urbanization. The region's harsh summers have led to air conditioning accounting for 70% of the annual peak electricity consumption in the residential sector, highlighting its dominance in energy usage (Global Carbon Council, 2025).

Low-Medium Enthalpy Geothermal Resources Modelling. Table 2 and Figure 11 represent modeling work that was performed for six scenarios (scenarios 1 to 6) in a low-medium enthalpy GCC-similar geology having a bottom-hole temperature of 150 °C and a range of three thermal conductivities to estimate the anticipated cooling capacity from a single closed-loop geothermal well, assuming a COP of 0.8, which is typical for a single-absorption (low-heat) chiller.

Table 2—Modeling for six scenarios for drilling a single closed-loop geothermal well in a low-medium enthalpy GCC-similar geology and thermal conditions.

Scenarios	#1	#2	#3	#4	#5	#6
Low-heat absorption chiller COP	0.8					
Thermal conductivity	2.0 W/(m·K)		3.5 W/(m·K)		5.5 W/(m·K)	
Reservoir temperature (Bottom-hole temperature)	150 °C					
Temperature at surface (outlet) / Return fluids temperature	80 °C/30 °C	100°C/30°C	80°C/30°C	100°C/30°C	80°C/30°C	100°C/30°C
10-year average heat extraction (MWth)	1.04	0.80	1.73	1.41	2.6	2.1
Cooling capacity (MWth)	0.83	0.64	1.38	1.13	2.07	1.7
Flow rate (kg/s)	4.5	2.4	7.6	4.4	11.4	6.6

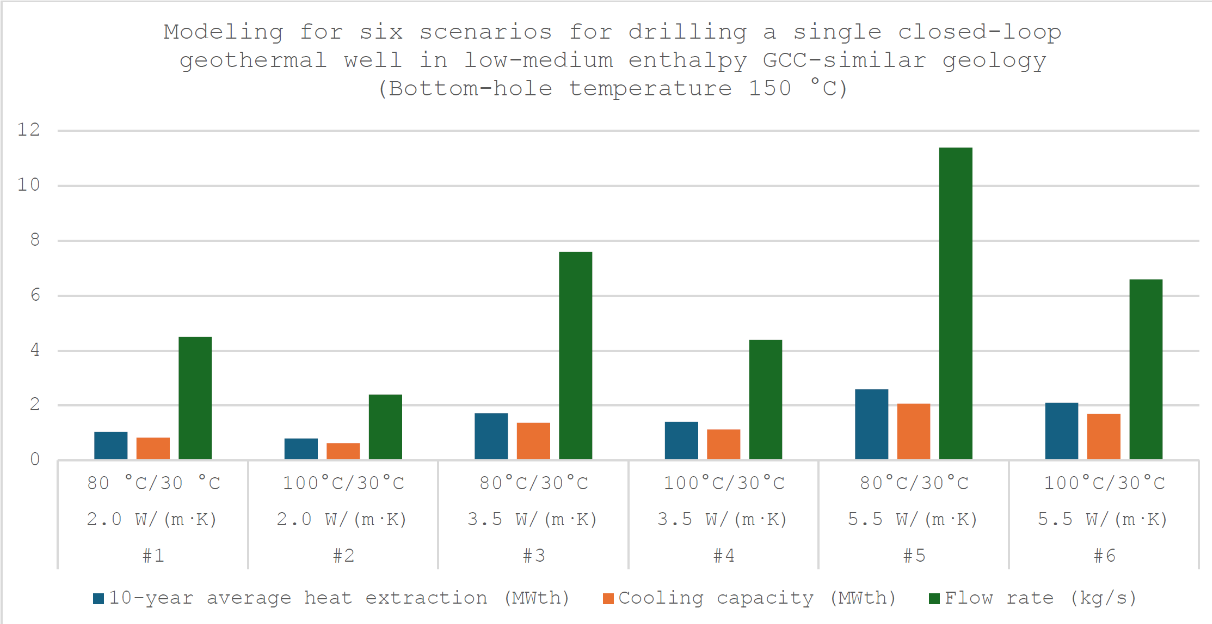

Figure 11—Graphical modeling for six scenarios for drilling a single closed-loop geothermal well in a low-medium enthalpy GCC-similar geology and thermal conditions.

The modeling was performed in a very conservative manner, assuming low, medium, and high thermal conductivities, which are found in the different geothermal reservoir lithologies, each having a different thermal conductivity, depending on the rock's petrophysical and geomechanical properties. The three chosen thermal conductivities for modeling were 2.0 W/(m·K) for low thermal conductivity rocks such as low-density clastics, 3.5 W/(m·K) for medium thermal conductivity rocks such as basement rocks or sandstone, and 5.5 W/(m·K) for high thermal conductivity rocks such as rock salt. Furthermore, depending on the length of the horizontal section, the bottom-hole temperature and the fluid flow rate, two outlet fluid temperatures were predicted for the fluids coming out of the well, with 80 °C being the low end and 100°C being the high end. One would expect that a higher outlet fluid temperature would always translate to higher heat, extraction, and cooling capacity, but in reality, in a closed-loop well, a higher outlet fluid temperature does not lead to higher heat extraction. This is because, in order to achieve a higher outlet fluid temperature (100 °C), the system should run at a lower fluid flow rate, to allow the fluid to have more time to heat up. But that lower flow will result in less total fluid being extracted at surface, leading to less efficient heat extraction. With a slightly cooler target outlet fluid temperature (80 °C), a higher fluid flow rate can be achieved, which moves more fluid through the well and improves heat transfer, leading to more total heat extracted and therefore more cooling capacity, even though the outlet temperature is lower. So, a very careful modeling for the fluid flow rate versus outlet fluid temperature is crucial for achieving the optimum scenario, which will produce the highest possible cooling capacity.

As can be observed from the modeling, 2.07 MWth cooling capacity is modelled as the best-case scenario, achieved with 5,5 W/(m·K) thermal conductivity and 80 °C outlet fluid temperature at a flow rate of 11.4 kg/s. The lowest predicted cooling capacity was 0.64 MWth, which was achieved with 2 W/(m·K) thermal conductivity and 100 °C outlet fluid temperature at a flow rate of 2.4 kg/s.

High Enthalpy Geothermal Resources Modelling. Table 3 – Modeling for six scenarios for drilling a single closed-loop geothermal well in a high enthalpy GCC-similar geology and thermal conditions. The second modeling work was done with one difference, which was changing the bottom hole temperature from 150 °C to 200 °C to model for high enthalpy GCC similar geology and evaluate the estimated produced cooling capacity in a high enthalpy environment. Table 3 and Figure 12 represent six scenarios which were modeled (scenarios 7 to 12), and as can be observed from Table 3, scenario 11 with a thermal conductivity of 5.5 W/(m·K) and outlet fluid temperature of 80 °C, estimated a cooling capacity of 3.4 MWth, which is 64.2% higher than the highest modeled cooling capacity in Table 1 (Scenario 5). So, increasing the bottom hole temperature from 150 °C to 200 °C, which represents a 33.3% increase, resulted in a 64.2% increase in the cooling capacity. The calculations in Tables 2 and 3 are for the initial 10-year average heat production, and they are expected to be maintained for 50+ years.

Table 3—Modeling for six scenarios for drilling a single closed-loop geothermal well in a high enthalpy GCC-similar geology and thermal conditions.

Scenarios	#7	#8	#9	#10	#11	#12
Low-heat absorption chiller COP	0.8					
Thermal conductivity	2.0 W/(m·K)		3.5 W/(m·K)		5.5 W/(m·K)	
Reservoir temperature (Bottom-hole temperature)	200 °C					
Temperature at surface (outlet) / Return fluids temperature	80°C/30 °C	100°C/30°C	80°C/30°C	100°C/30°C	80°C/30°C	100°C/30°C
10-year average heat extraction (MWth)	1.7	1.5	2.9	2.6	4.3	3.9
Cooling capacity (MWth)	1.4	1.2	2.3	2.1	3.4	3.1
Flow Rate (kg/s)	7.5	4.7	12.6	8.0	18.7	12.0

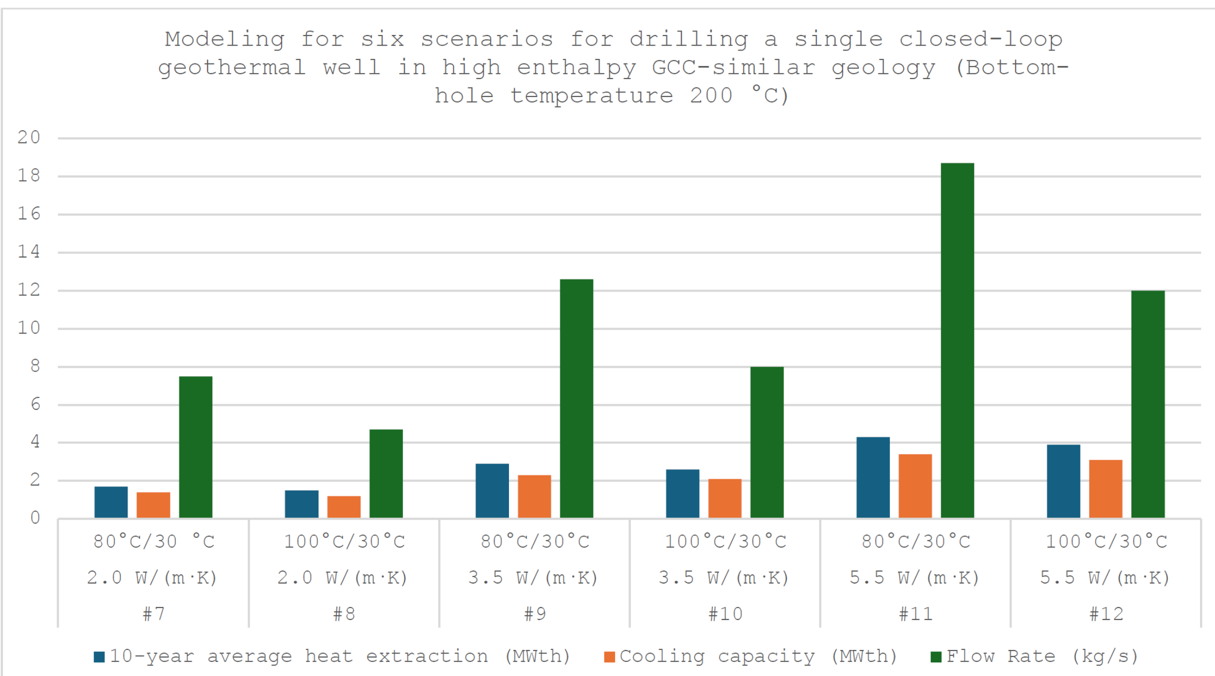

Figure 12—Graphical modeling for six scenarios for drilling a single closed-loop geothermal well in a high enthalpy GCC-similar geology and thermal conditions.

Discussion & Conclusions

As the urgency of transition to more sustainable energy sources grows, driven mainly by the rising frequency of natural disasters worldwide, numerous challenges still hinder the widespread adoption of environmentally friendly energy alternatives. While renewable and low-emission energy options such as wind, solar, hydro, and geothermal offer clear environmental benefits, their adoption is often constrained by a combination of factors. These include high initial capital costs, the need for significant upgrades to existing energy infrastructure, intermittency issues with certain renewable sources, and the lack of robust energy storage solutions. Additionally, regulatory hurdles, inconsistent policy support, and the need for skilled labor and technological innovation present further obstacles. Public resistance, land use conflicts, and concerns about environmental impacts from large-scale renewable projects can also slow progress. Despite these challenges, the transition is essential to mitigating future climate risks and ensuring a resilient, low-carbon energy future.

Geothermal energy, when compared with solar and wind energy, offers key advantages over solar and wind, mainly due to its reliability and efficiency. Unlike solar and wind, which are intermittent and weather-dependent, geothermal provides a constant 24/7 energy supply, making it a reliable baseload energy source without the need for large-scale storage. It also requires a much smaller surface footprint, with most infrastructure located underground, resulting in minimal visual and environmental impact. Additionally, geothermal systems have higher capacity factors and longer operational lifespans, often exceeding 50 years, with relatively low maintenance and operational costs. While solar and wind are valuable renewable sources, geothermal stands out for its consistency, space efficiency, and long-term performance where geological conditions allow.

Despite global cooling demand being higher than heating, the market for geothermal district cooling remains less developed compared to district heating. However, in the GCC region, investment in district cooling is rising steadily, with plans for widespread implementation particularly in UAE and Saudi Arabia. Consequently, the district cooling sector is experiencing rapid growth and is projected to expand even further in the coming years.

Harnessing the abundant geothermal energy for district cooling can be a cornerstone in decarbonizing energy usage and making electricity available for many other applications. With the hydrothermal doublet solution, it has not been possible to scale up the usage of geothermal heat for district cooling. However, with the single closed-loop geothermal well solution, that is now a possibility.

For district cooling applications, a single closed-loop well, when compared to the doublet well option, offers several advantages:

- Requires only one well to be operational
- It has fewer geological requirements, mainly thermal conductivity of the rock, and the geothermal temperature gradient, which means that geological requirements for this solution can be easily identified around the globe. A geothermal district cooling can therefore be more widespread with the implementation of this solution.
- Reduced risk of project failure, because the single closed-loop well is guaranteed to generate thermal energy, and the only question that remains is how much.
- Reduced and almost eliminated the need for well maintenance, interventions, or stimulation, which significantly brings down the operational expenditures and reduces the risk of operational failures or shutdowns.
- Due to its increased reliability as a solution, the lifespan of this option can be 50 years or longer, which is much longer than the average lifespan of the doublet well option, which has a lifespan of 30 years on average.

Therefore, single closed-loop well technology offers a great advantage by significantly reducing both the CAPEX requirements to develop district cooling projects and the operating expenditure. It also increases the district cooling project lifespan when compared to the doublet hydrothermal well solution, making it a more economically feasible and practical solution. When coupled with an absorption chiller system on surface, this configuration offers a very economically feasible, energy-efficient, and reliable district cooling solution.

Acknowledgment

Thanks to TAQA, Green Therma, and the University of Exeter for allowing the use of their technology, data, and simulation software in this research.

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